

Decision trees are a popular method for various machine learning tasks. Tree learning is almost "an off-the-shelf procedure for data mining", say Hastie et al., "because it is invariant under scaling and various other transformations of feature values, is robust to inclusion of irrelevant features, and produces inspectable models. However, they are seldom accurate". In particular, trees that are grown very deep tend to learn highly irregular patterns: they overfit their training sets, i.e. have low bias, but very high variance. Random forests are a way of averaging multiple deep decision trees, trained on different parts of the same training set, with the goal of reducing the variance. This comes at the expense of a small increase in the bias and some loss of interpretability, but generally greatly boosts the performance in the final model.

ExtraTrees Adding one further step of randomization yields extremely randomized trees, or ExtraTrees. As with ordinary random forests, they are an ensemble of individual trees, but there are two main differences: (1) each tree is trained using the whole learning sample (rather than a bootstrap sample), and (2) the top-down splitting is randomized: for each feature under consideration, a number of random cut-points are selected, instead of computing the locally optimal cut-point (based on, e.g., information gain or the Gini impurity). The values are chosen from a uniform distribution within the feature's empirical range (in the tree's training set). Then, of all the randomly chosen splits, the split that yields the highest score is chosen to split the node. Similar to ordinary random forests, the number of randomly selected features to be considered at each node can be specified. Default values for this parameter are $\sqrt{p}$ for classification and p for regression, where p is the number of features in the model.

Variants Instead of decision trees, linear models have been proposed and evaluated as base estimators in random forests, in particular multinomial logistic regression and naive Bayes classifiers. In cases that the relationship between the predictors and the target variable is linear, the base learners may have an equally high accuracy as the ensemble learner.

Variable importance Random forests can be used to rank the importance of variables in a regression or classification problem in a natural way. The following technique was

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Prefiltering: Eliminate features that are mostly just noise.
Enriched Random Forest (ERF): Use weighted random sampling instead of simple random sampling at each node of each tree, giving greater weight to features that appear to be more informative. Tree-weighted random forest (TWRF): Give more weight to more accurate trees.

Unsupervised learning As part of their construction, random forest predictors naturally lead to a dissimilarity measure among observations. One can analogously define dissimilarity between unlabeled data, by training a forest to distinguish original "observed" data from suitably generated synthetic data drawn from a reference distribution. A random forest dissimilarity is attractive because it handles mixed variable types very well, is invariant to monotonic transformations of the input variables, and is robust to outlying observations. Random forest dissimilarity easily deals with a large number of semi-continuous variables due to its intrinsic variable selection; for example, the "Addcl 1" random forest dissimilarity weighs the contribution of each variable according to how dependent it is on other variables. Random forest dissimilarity has been used in a variety of applications, e.g. to find clusters of patients based on tissue marker data.

Consistency of uniform KeRF Providing $k \rightarrow \infty$ and $n/2^k \rightarrow \infty$, there exists a constant $C > 0$ such that,

Random forests for high-dimensional data The basic random forest procedure may not work well in situations where there are a large number of features but only a small proportion of these features are informative with respect to sample classification. This can be addressed by encouraging the procedure to focus mainly on features and trees that are informative. Some methods for accomplishing this are:

$$K_{k \cup \{f\}}(\mathbf{0}, \mathbf{x}) = \sum_{k_1=0}^{k_1} \dots \sum_{k_d=0}^{k_d} \sum_{j=1}^d k_j = k! k_1! \dots k_d! (1-d) k \prod_{m=1}^d (1 - |x_m| \sum_{j=0}^{k_m-1} (-\ln |x_m|)^j j!) \text{ for all } \mathbf{x} \in [0, 1]^d.$$

$$\{x\} \in [0,1]^d.$$

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Mean decrease in impurity feature importance This approach to feature importance for random forests considers as important the variables which decrease a lot the impurity during splitting. It is described in the book Classification and Regression Trees by Leo Breiman and is the default implementation in sci-kit learn and R. The definition is: unnormalized average importance $(x) = \frac{1}{n} \sum_{i=1}^n \sum_{\text{node } j \in T_i} \text{split variable}(j) = x \cdot p_{T_i}(j) \Delta i_{T_i}(j)$,
$$\text{unnormalized average importance}(x) = \frac{1}{n} \sum_{i=1}^n \sum_{\text{node } j \in T_i} p_{T_i}(j) \Delta i_{T_i}(j),$$
 where

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From random forest to KeRF Given a training sample $D_n = \{ (X_i, Y_i) \}_{i=1}^n$ of $[0, 1]^p \times \mathbb{R}$ -valued independent random variables distributed as the independent prototype pair (X, Y) , where $E[Y^2] < \infty$. We aim at predicting the response Y , associated with the random variable X , by estimating the regression function $m(x) = E[Y | X = x]$. A random regression forest is an ensemble of M randomized regression trees. Denote $m_n(x, \Theta_j)$ the predicted value at point x by the j -th tree, where $\Theta_1, \dots, \Theta_M$ are independent random variables, distributed as a generic random variable Θ , independent of the sample D_n . This random variable can be used to describe the randomness induced by node splitting and the sampling procedure for tree construction. The trees are combined to form the finite forest estimate $m_{M,n}(x, \Theta_1, \dots, \Theta_M) = \frac{1}{M} \sum_{j=1}^M m_n(x, \Theta_j)$. For regression trees, we have $m_n = \sum_{i=1}^n Y_i \mathbb{1}_{X_i \in A_n(x, \Theta_j)}$, where $A_n(x, \Theta_j)$ is the cell containing x , designed with randomness Θ_j and dataset D_n , and $N_n(x, \Theta_j) = \sum_{i=1}^n \mathbb{1}_{X_i \in A_n(x, \Theta_j)}$. Thus

random forest estimates satisfy, for all $x \in [0, 1]^d$
$$m_{M,n}(x, \Theta_1, \dots, \Theta_M) = \frac{1}{M} \sum_{j=1}^M \left(\sum_{i=1}^n \frac{Y_i}{N_n(x, \Theta_j)} \mathbb{1}_{X_i \in A_n(x, \Theta_j)} \right).$$
 Random regression forest has two levels of averaging, first over the samples in the target cell of a tree, then over all trees. Thus the contributions of observations that are in cells with a high density of data points are smaller than that of observations which belong to less populated cells. In order to improve the random forest methods and compensate the misestimation, Scornet defined KeRF by